LAMIA ISLAM

BANGLADESH

RELIABLE ML · OUT-OF-DISTRIBUTION DETECTION · EVALUATION VALIDITY

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RESEARCH INTERESTS

Information and Communication Engineering graduate from Pabna University of Science and Technology, interested in reliable machine learning, multi-agent systems, and autonomous decision-making under environmental uncertainty. My research experience spans out-of-distribution detection, physiological signal monitoring, and multi-robot search and rescue. I developed a collaborative search-and-rescue system using ROS 2 and Gazebo, where I investigated the limitations of shared probabilistic beliefs in heterogeneous environments and explored path-aware alternatives. I am particularly interested in how autonomous agents can share information, learn collaboratively, and make reliable decisions when their observations and operating environments differ.

EDUCATION

Pabna University of Science and Technology (PUST), Bangladesh Feb 2022 – Aug 2026
B.Sc. (Engineering) in Information and Communication Engineering

- CGPA 3.34/4.00; final two semesters 3.77/4.00.
- Thesis: *Design and Evaluation of a Competence-Gated Driver-State Monitoring System for Autonomous Intervention Control*. Supervisor: Prof. Dr. Md. Anwar Hossain. Defended 2 September 2026.

Higher Secondary Certificate (HSC), Science · *GPA 5.00/5.00* 2020
Secondary School Certificate (SSC), Science · *GPA 5.00/5.00* 2018

ENGLISH PROFICIENCY

IELTS Academic Nov 2026
Overall **7.5** (CEFR C1) · Listening 7.5 · Reading 8.5 · Writing 6.5 · Speaking 7.5

RESEARCH EXPERIENCE

Research Assistant, Department of ICE, PUST 2026 – Present
Supervisor: Prof. Dr. Md. Anwar Hossain

- Built a density-based (class-conditional Gaussian) classifier of physiological stress from ECG and electrodermal activity. Balanced accuracy 0.84 on WESAD (15 participants, leave-one-subject-out).
- Tested it on real driving recordings (MIT DriveDB) from unseen people. Posterior-based confidence scores ranked the driving data as more familiar than the training data (AUROC 0.36 to 0.46). The model's marginal log-evidence, a value it already computes and normally discards, ranked it correctly (AUROC 0.77 to 0.84) at no extra cost.
- Showed that how well a detector ranks unfamiliar data does not predict how it behaves as a gate on intervention. The best-ranked score held back escalation in 96% of driving windows, against 54% for the classifier's own score.
- Caught a scale artifact between datasets that inflated detection before reporting, and found participants poorly described by the model even within its training data.
- Deployed the monitor in ROS 2 Humble and Gazebo as a graded, human-overridable intervention (alert, slow, stop). Sole-author manuscript under review [J2].

Research Assistant, Department of ICE, PUST 2024 – Mar 2026
Supervisor: Dr. Md. Imran Hossain

- Carried three projects from formulation to publication: nightmare prediction in children, hepatitis C detection, and an RFID access-control system with networked logging [C1, C2, C4].

PUBLICATIONS

JOURNAL MANUSCRIPTS UNDER REVIEW

[J1] **L. Islam** (first and corresponding author), M. Einollahi Asgarabad. Site identity leaks through missingness: unsupervised competence monitoring under-detects shift in four cardiovascular cohorts. *Royal Society Open Science*, submitted Oct 2026. Code and audit: doi.org/10.5281/zenodo.23205549

[J2] **L. Islam** (sole author). Thresholds set on training data understate the false alarms of competence gates for a wearable driver-state monitor. *SN Computer Science* (Springer), submitted Oct 2026.

[J3] **L. Islam** (first and corresponding author), M. S. Hosain. Synthetic benchmark validity in medical image segmentation. *Journal of Imaging Informatics in Medicine* (Springer), submitted Oct 2026.

[J4] M. B. A. Z. Shammo, T. Debnath, **L. Islam**, S. Sarker, J. Habib. Vision-based crack segmentation and coverage planning for autonomous mobile robots. *Int. J. of Intelligent Robotics and Applications* (Springer), revision submitted Sep 2026. Preprint: doi.org/10.21203/rs.3.rs-9996410/v1

PEER-REVIEWED CONFERENCE PAPERS

[C1] **L. Islam**, M. B. A. Z. Shammo, M. I. Hossain. Real-time RFID authentication and access control system with networked logging. *IEEE QPAIN*, Chattogram, Apr 2026. doi.org/10.1109/QPAIN69676.2026.11546467

[C2] M. B. A. Z. Shammo, **L. Islam**, M. Hasan, M. A. Hossain, M. I. Hossain, M. A. Mia, M. H. Mohona. Machine learning-based hepatitis C detection using ensemble boosting. *3rd Int. Conf. on Big Data, IoT and Machine Learning*, Springer Nature, Apr 2026. doi.org/10.1007/978-3-032-15764-5_50

[C3] M. A. Miya, M. H. Mohona, M. A. Hossain, M. B. A. Z. Shammo, A. Islam, **L. Islam**. Remote health monitoring via PPG signal processing and machine learning using mobile devices. *IEEE ICCIT*, Cox's Bazar, Dec 2025. doi.org/10.1109/ICCIT68739.2025.11491546

[C4] **L. Islam**, M. B. A. Z. Shammo, M. Rashed, M. I. Hossain, M. A. Miya, M. H. Mohona. Machine learning-based prediction of nightmares in children: a pediatric psychological integration. *IEEE COMPAS*, Kushtia, 2025. doi.org/10.1109/COMPAS67506.2025.11381639

RESEARCH PROJECTS

Multi-Robot Search and Rescue 2026

ROS 2 Humble, Gazebo Fortress · Submitted to IEEE ICCIT 2026

- Three differential-drive robots map a collapsed building together: slam_toolbox mapping, a custom map-merge node, a wavefront BFS coordinator with Hungarian frontier assignment, a pure-pursuit controller, and a shared Gaussian belief map of where survivors are likely to be.
- Found that this belief helps in open space but misleads robots in rubble, because it spreads through walls. A path-aware (geodesic) version partly corrects it. Code: github.com/Lamia-Islam

When Benchmarks Mislead: Medical Image Segmentation 2026

57 methods, 11 clinical datasets in four modalities · Under review [J3]

- Scored 57 segmentation methods, including ImageNet-pretrained architectures and nnU-Net, on 11 public clinical datasets and on synthetic wound benchmarks.
- Showed that rank agreement between clinical datasets is organised by modality, and that pooling untrained with trained methods manufactures agreement (0.13 rising to 0.80), so a benchmark's measured validity depends on the methods it is tested with.

SKILLS

Machine learning	Python, PyTorch, TensorFlow, scikit-learn; out-of-distribution detection, calibration, leave-one-subject-out and leave-one-cohort-out validation, statistical testing
Robotics	ROS 2 Humble, Gazebo Fortress, slam_toolbox, MoveIt 2, RViz, Ubuntu (WSL2)
Signals and hardware	ECG, EDA and PPG processing, HRV features; ESP32, AD8232, GSR sensors, Arduino, RFID
Other	C, R, MATLAB, Git, LaTeX

SERVICE AND LEADERSHIP

Robotics Secretary, ICE Association, PUST Oct 2025 – 2026

Peer Reviewer, IEEE Conferences: IEEE RAAICON 2026; IEEE RAAICON 2025 2025 – 2026

REFERENCES

Prof. Dr. Md. Anwar Hossain

Professor and Chairman, Department of ICE, and Dean, Faculty of Engineering and Technology, PUST. Thesis supervisor.

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Prof. Dr. Md. Sarwar Hosain

Professor, Department of ICE, PUST. Co-author, medical image segmentation manuscript [J3].

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Dr. Md. Imran Hossain

Associate Professor, Department of ICE, PUST. Research mentor, 2024 to 2026.

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